CALORIES TRACKER
ML MINI PROJECT

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Machine Learning for Calories-Burned Estimation and FoodNutrition Tracking

Prepared for academic project evaluation

Based on the submitted notebook, exported model artifacts, and UI recording

1. Introduction

Fitness applications are most useful when they turn everyday activity information into estimates that people can understand. In practice, calories burned during a workout depend on several factors, including how long the activity lasts, the person's body measurements, heart rate, and exercise intensity. A fixed rule can be helpful as a rough guide, but it may not reflect every individual or workout equally well.

This project, Calories Tracker, explores a data-driven alternative. It trains regression models to estimate calories burned from activity and physiological features, then places the selected model inside a simple interactive interface. A separate food database supports calorie lookup for foods, while BMI and daily energy needs are calculated using formulas. These functions are related, but they should not be confused: the machine-learning model estimates *energy expenditure*, while the food calculator estimates *energy intake* from nutrition data.

1.1 Problem statement

People often rely on generalized calorie estimates that may not match their workout conditions. The project investigates whether structured activity data can be used to predict calories burned and how the result can be presented through a user-friendly interface. A second practical challenge is data quality: missing values, duplicate rows, inconsistent types, and unsuitable features can weaken a model before training even begins.

1.2 Objectives

- Clean and validate the activity/BMI dataset and the food nutrition dataset.
- Prepare a reproducible training and testing workflow.
- Compare at least four regression approaches using MAE, RMSE, and R^2.
- Select a model based on held-out test performance.
- Provide a UI for calories-burned estimation and food-calorie lookup.
- Include BMI and daily energy estimates with clear educational limitations.

1.3 Scope and responsible use

The current implementation is an educational estimator, not a medical device or a substitute for laboratory measurement. The exported test results describe performance on the dataset split used in the notebook; they do not guarantee the same accuracy for every person, sensor, or activity type. Users should treat the prediction as an estimate rather than an exact measurement.

2. Literature Review

Research on energy-expenditure estimation increasingly combines physiological signals, movement sensors, and machine learning. The studies below are relevant because they investigate the same broad question as Calories Tracker: how to estimate energy use from measurable activity and personal characteristics. Their datasets and evaluation settings differ from the project dataset, so their metrics should not be compared as if they came from one shared benchmark.

2.1 Jeran et al. (2022): activity-related expenditure in free-living adults

Jeran and colleagues developed prediction models for activity-related energy expenditure using accelerometer-derived activity and additional participant information. Their work highlights that movement data alone may not explain all variation; body composition and lifestyle-related variables can add useful information. The study is valuable for this project because it reinforces the importance of choosing features that represent both the activity and the individual. Unlike this notebook, the study considered free-living energy expenditure and accelerometry-derived variables rather than the project's structured workout table. [1]

2.2 "Burned Calories Prediction using Supervised Machine Learning" (2022)

This IEEE conference paper compares supervised regression approaches for predicting calories burned and reports Random Forest as its strongest model in its experimental setup. It supports the use of tree-based ensembles as sensible benchmarks for this problem. The result is not proof that Random Forest must win on every dataset; model rankings can change with feature definitions, data collection, and validation design. Calories Tracker therefore compares Random Forest with several alternatives on its own held-out test set. [2]

2.3 Dynamic personalized energy expenditure with LSTM models (2023)

A 2023 study explored time-series approaches, including LSTM, CNN-LSTM, and an ensemble combining sequence modelling with gradient boosting. The authors emphasized that energy expenditure changes over time and that personalized models may capture patterns that a generalized model misses. This is an important direction for future work. The current Calories Tracker model uses row-level structured features rather than continuous sensor sequences, so it should not be described as a dynamic time-series predictor. [3]

2.4 What these studies imply

Across these papers, a consistent message is that feature quality, measurement context, and validation strategy matter alongside the algorithm itself. They also caution against assuming that one model or one metric tells the whole story. Calories Tracker follows this principle by comparing several regressors on the same test split and reporting error in kcal as well as R^2.

2. Literature Review — Comparison and Research Gap

2.5 Recent studies on wearable sensing and energy expenditure

Lendt et al. (2024) proposed a composite model that combines activity-type classification with stride-specific energy-expenditure estimation from thigh-worn accelerometry. Their reported results show that distinguishing walking, running, and cycling can be useful because the relationship between movement and energy cost differs by activity. Calories Tracker does not currently classify activity type, but this work suggests a practical future extension. [4]

The 2024 CalAid paper describes a mobile fitness application that uses smartphone motion sensors, activity recognition, and metabolic-equivalent information to estimate energy expenditure. Its contribution is not only a model but also the integration of sensing and user-facing application functionality. This is closely related to the present project's goal of making a model usable through an interface, although Calories Tracker currently uses user-entered structured features rather than a continuous smartphone-sensor stream. [5]

Brønd (2024) investigated LSTM recurrent neural networks for energy-expenditure prediction from accelerometry data. The study supports sequence-based modelling as a promising option when observations have temporal structure. For the current

project, such models would require appropriate time-series inputs and validation that respects participants or sessions; simply applying an LSTM to tabular rows would not automatically improve reliability. [6]

2.6 Comparison of literature

Study	Year	Approach / data	Main relevance to Calories Tracker
Jeran et al. [1]	2022	Accelerometry + participant factors	Use meaningful activity and personal features
IEEE supervised ML [2]	2022	Regression; Random Forest reported	estInclude tree ensembles in the benchmark
Dynamic personalized	E [3]2023	LSTM / CNN-LSTM / ensemble	Temporal and personalized modelling may help
Lendt et al. [4]	2024	Activity recognition + stride-specific mo	delActivity type can affect energy estimates
CalAid [5]	2024	Mobile app, phone sensors, activity re	ognitionConnect estimation to usable interface
Brønd [6]	2024	LSTM using accelerometry	Time-series models are a future option

2.7 Research gap and project positioning

The reviewed work often relies on wearable sensors, activity sequences, or carefully measured laboratory/free-living outcomes. The submitted project instead demonstrates an end-to-end tabular workflow: cleaning, feature preparation, model comparison, saved model artifacts, and an interactive calorie tracker. Its contribution is an accessible implementation and a transparent benchmark on the available dataset—not a claim of outperforming research-grade wearable systems. A key limitation is that generalization to new people and devices has not been established by the supplied notebook.

3. Methodology

The implementation follows a practical machine-learning pipeline. The notebook loads two CSV datasets, audits their structure, cleans them independently, defines the target, prepares the predictors, splits the activity data into training and test sets, fits multiple regression models, compares their test metrics, and saves the selected model and supporting files.

3.1 Data sources and target

The activity/BMI dataset initially contains 15,020 rows and 10 columns. After exact duplicate removal and validation, it contains 15,000 rows; 14,992 rows have a known target value for training. The target is **Calories_Burned_kcal**. The food nutrition dataset initially contains 2,035 rows and 8 columns and is reduced to 2,025 rows after duplicate removal and cleaning.

3.2 Input features

The notebook uses eight predictors: Gender, Age_years, Height_cm, Weight_kg, Duration_min, Heart_Rate_bpm, Body_Temp_C, and BMI. Seven are numeric and Gender is categorical. This feature set is plausible for estimating energy expenditure because it includes workout duration, physiological response, and basic body measurements. It does not include a directly measured activity label or continuous accelerometer stream.

3.3 Training and evaluation design

The data is split into 80% training and 20% test data with random_state = 42. The resulting split contains 11,993 training rows and 2,999 test rows. The notebook uses a preprocessing-and-model pipeline, including imputation and encoding for the relevant feature types. Keeping preprocessing in the fitted pipeline helps apply the same transformations at training and prediction time.

3.4 Model selection

The notebook compares a mean Dummy Regressor baseline with Linear Regression, Ridge Regression, Decision Tree, Random Forest, Extra Trees, and Gradient Boosting. The best model is selected by the lowest test RMSE in the saved comparison output. Gradient Boosting is selected for the submitted run. For a more rigorous final study, hyperparameter tuning should be conducted only within training data using cross-validation, while the test set is reserved for a final unbiased estimate.

3.5 Evaluation metrics

Metric	Meaning	How to read it
MAE (kcal)	Average absolute difference between actual and predicte	d caloriesLower is better; easy to interpret in kcal
RMSE (kcal)	Square root of average squared prediction error	Lower is better; penalizes large errors more strongly
R^2	Proportion of target variation explained relative to a mean	baselineCloser to 1 is better; can be negative

4. Data Cleaning and Preprocessing

Data preparation is a major part of this project because the model cannot recover reliable patterns from inconsistent inputs. The notebook applies cleaning steps to both datasets before the model workflow begins.

4.1 Initial audit and duplicate removal

The audit checks dataset dimensions, data types, missing values, and exact duplicate rows. Twenty duplicate rows are identified in the activity/BMI dataset and five in the food dataset. Removing exact duplicates leaves 15,000 activity records and 2,025 food records. This prevents identical records from being counted multiple times, although duplicate removal does not by itself detect every possible data-quality issue.

4.2 Type conversion and validation

Column names are stripped of surrounding whitespace. Expected numeric fields are converted carefully so that invalid text does not silently remain as a numeric value. The target is checked before modelling, and rows without a known target are excluded from the supervised training data. The resulting model input is kept separate from the target to reduce the risk of accidentally including the answer as a predictor.

4.3 Feature processing

Numeric and categorical columns are identified separately. Numeric features are handled through an imputation path, while Gender is treated as categorical and encoded for model use. This is preferable to assigning arbitrary numeric codes to categories because the model should not infer a false numeric distance between gender labels. The exact transformations are encapsulated in the notebook's pipeline and saved with the estimator.

4.4 Food database cleaning

The food nutrition table is cleaned independently. It contains 2,025 items after preprocessing. The food lookup uses the foodname field and the calories-per-100-gram field; the estimated calories for a logged quantity are calculated proportionally. For example, a 150 g serving uses 1.5 times the listed per-100 g value. This assumes the database value is representative of the selected food and serving.

4.5 Potential data-quality risks

The notebook removes exact duplicates and handles types, but real datasets can still contain impossible physiological values, unit inconsistencies, mislabeled records, or unusual outliers. The report does not claim these issues have all been eliminated. Before wider deployment, ranges should be checked against domain rules, units should be documented, and missingness patterns should be reviewed. If multiple records belong to the same person, a random row split may also allow person-specific patterns to appear in both training and testing; a person-level split would better assess generalization to unseen users.

4.6 Reproducibility and saved outputs

The exported artifact bundle includes the selected model, cleaned activity/BMI data, cleaned food data, model comparison table, and metadata JSON. These files make the experiment easier to inspect and help the application reuse the fitted model without retraining on every request.

5. Results and Discussion

5.1 Dataset summary

Item	Reported value
Activity/BMI data before cleaning	15,020 rows × 10 columns
Activity/BMI data after cleaning	15,000 rows × 10 columns
Rows with known target	14,992
Food nutrition data before cleaning	2,035 rows × 8 columns
Food nutrition data after cleaning	2,025 rows × 8 columns
Train / test rows	11,993 / 2,999
Selected target	Calories_Burned_kcal

5.2 Selected model performance

Gradient Boosting is the best-performing model in the supplied results, with MAE = 20.453 kcal, RMSE = 25.510 kcal, and R^2 = 0.9711. In plain language, the model's average absolute prediction error on the test set is about 20.5 kcal, and the R^2 indicates that it explains a large share of the variation in the held-out target values under this split. RMSE is higher than MAE, which suggests some larger errors have more influence on the squared-error measure.

Selected-model metric	Result	Interpretation
MAE	20.453 kcal	Typical absolute error scale
RMSE	25.510 kcal	Sensitive to larger misses
R^2	0.9711	Strong fit on this test split

5.3 Comparison against the simple baseline

The Dummy Regressor has MAE of 129.412 kcal, RMSE of 150.163 kcal, and R^2 slightly below zero. Gradient Boosting performs substantially better than this mean-prediction baseline on the supplied test split. That is encouraging, but a high R^2 alone does not establish clinical accuracy or performance on unseen populations. The target-generation process, sampling design, and potential similarity between training and test records can influence the apparent score.

5.4 Interpretation

The results support the project's core objective: a trained regression model can produce useful estimates from the available tabular features. Tree-based ensembles perform particularly well in this experiment, while Linear and Ridge Regression also perform strongly. A practical next step is error analysis by workout duration, heart-rate range, age group, and body measurements to identify where predictions are less reliable. If the dataset contains participant identifiers, evaluation should also be repeated with a participant-separated split.

6. Benchmark Model Comparison

The comparison below uses the saved test-set metrics from the submitted artifact bundle. All models were evaluated on the same target and split in that run. MAE and RMSE are expressed in kcal; lower values indicate smaller errors. R^2 is unitless and higher values are preferable.

Model	MAE (kcal)	RMSE (kcal)	R^2
Gradient Boosting	20.453	25.510	0.9711
Random Forest	20.934	26.229	0.9695
Extra Trees	21.125	26.389	0.9691
Linear Regression	23.064	29.225	0.9621
Ridge Regression	23.064	29.225	0.9621
Decision Tree	26.296	32.958	0.9518
Dummy baseline	129.412	150.163	-0.0003

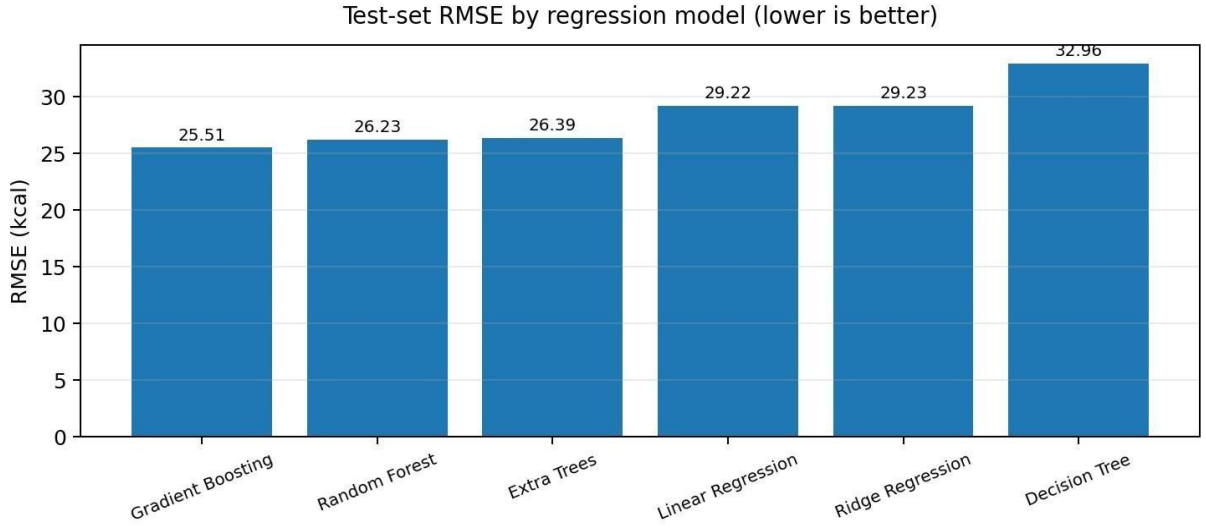


Figure 1. RMSE comparison using the project's saved test-set results. Gradient Boosting has the lowest RMSE among the tested regressors.

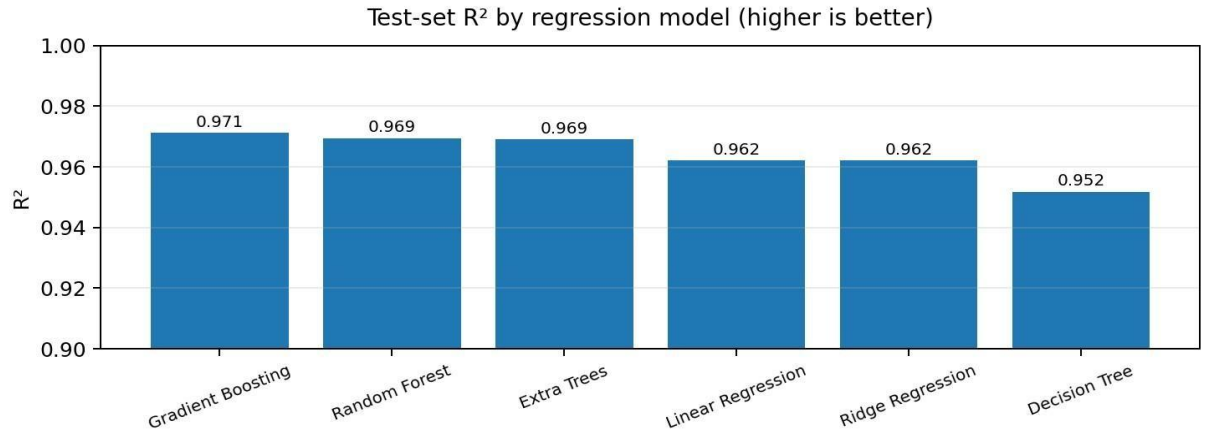


Figure 2. R^2 comparison for the six trained regressors. The Dummy baseline is omitted from the chart because its near-zero R^2 would compress the scale; its exact value is included in the table.

6.1 Model-by-model discussion

Gradient Boosting ranks first by RMSE, followed closely by Random Forest and Extra Trees. Linear Regression and Ridge Regression show very similar scores, suggesting that the regularization used by Ridge made little difference in this run. Decision Tree performs less strongly than the ensembles, which is consistent with the possibility that combining many trees improves stability. These are observations about this dataset and split, not universal rankings of the algorithms.

7. User Interface Development and Deployment

The supplied screen recording shows a dark-themed Calories Tracker interface running locally in a browser. The visible page is designed around a direct task: enter the user and workout attributes, calculate BMI, and request a calories-burned estimate. The layout uses grouped input controls and a clearly highlighted result area, which keeps the central output easy to notice.

7.1 Main interface behaviour

The notebook also includes a food logging / nutrition calculator, a formula-based daily calorie-needs estimate, and a BMI calculator. The calorie-burn model predicts expenditure; the food database estimates intake from food selection and quantity. The daily energy estimate is formula-based rather than machine-learning output, and BMI is a screening calculation based on height and weight.

7.2 Deployment considerations

The notebook exports a joblib model artifact and cleaned data files. A deployment should load the fitted preprocessing-andmodel pipeline, validate user inputs, handle missing or out-of-range values, and show prediction units consistently. The supplied recording demonstrates local UI operation; it does not establish that a public cloud deployment is live. Before public release, the project should be tested in a clean environment with pinned dependencies, artifact paths verified, and sensitive user data kept out of logs.

7.3 Usability, privacy, and limitations

A clear interface should label inputs, state that the output is an estimate, and avoid presenting calorie values as medically exact. Because body measurements and activity data can be personal information, a production version should explain how data is stored and avoid retaining user inputs unless necessary. Further improvements could include responsive mobile layout, prediction history with explicit consent, input-range warnings, uncertainty estimates, and an explanation of the most influential features.

7.4 Future scope

Future work could add participant-level validation, external testing, activity-type information, wearable sensor streams, timeseries models, and calibration for individual users. The nutrition module could be expanded with serving-size guidance and clear food-source provenance. These extensions should be evaluated separately so that interface improvements are not mistaken for gains in model accuracy.

8. Conclusion

Calories Tracker brings together data cleaning, regression benchmarking, model export, and a user-facing calorie estimation workflow. On the supplied test split, Gradient Boosting achieved the lowest RMSE (25.510 kcal), MAE of 20.453 kcal, and R^2 of 0.9711. Random Forest and Extra Trees were close alternatives, while the simple mean baseline performed substantially worse. The results demonstrate the value of comparing multiple models instead of choosing one by intuition.

The project also separates three tasks that are easy to confuse: calories burned are estimated by the machine-learning model, food calories are calculated from the nutrition database, and daily energy needs and BMI are formula-based outputs. This distinction improves transparency for users. The current implementation is a promising educational prototype, but it still needs external validation, participant-separated evaluation, stronger input validation, and careful deployment before it can be relied upon beyond the dataset used here.

9. References

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